

Sudakov - Fernique Comparison Inequality

Assume that $\underline{X} = (X_1, \dots, X_N)$ & $\underline{Y} = (Y_1, \dots, Y_N)$ both joint Gaussian, and zero-mean & $\underline{X} \perp \underline{Y}$, &

$$\forall i, j \quad : \quad \mathbb{E} \{ (X_i - X_j)^2 \} \leq \mathbb{E} \{ (Y_i - Y_j)^2 \},$$

$$\text{then} \quad \mathbb{E} \left\{ \max_{1 \leq i \leq N} X_i \right\} \leq \mathbb{E} \left\{ \max_{1 \leq i \leq N} Y_i \right\}.$$

Proof:

$$Z \sim \mathcal{N}(0, \sigma^2) \quad \mathbb{E} \{ Z f(Z) \} = \int_{-\infty}^{+\infty} z f(z) \phi(z) dz$$

$$\phi(z) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{z^2}{2\sigma^2} \right\}$$

$$\phi'(z) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{z^2}{2\sigma^2} \right\} \cdot \left(-\frac{z}{\sigma^2} \right)$$

$$= [z \phi(z)] \left[-\frac{1}{\sigma^2} \right]$$

$$\Rightarrow \mathbb{E} \{ z f(z) \} = \int_{-\infty}^{+\infty} \underbrace{-\sigma^2 \phi'(z)}_{dv} \underbrace{f(z)}_u dz$$

$$= -\sigma^2 \left[\underbrace{f(z) \phi(z)}_{=0 \text{ if } f \text{ doesn't grow faster than } 1/\phi} \right]_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} f'(z) \phi(z) dz$$

= 0 if f doesn't grow faster than $1/\phi$ in $-\infty$ & $+\infty$.

$$= \sigma^2 \int_{-\infty}^{\infty} f'(z) \phi(z) dz$$

$$= \sigma^2 \mathbb{E} \{ f'(z) \}.$$

(Gaussian Integration by part).

$$\mathbb{E} \{ W f(z) \}$$

assume W & z are joint Gaussian

Corr. Coeff. & zero-mean

$$W = \widehat{\rho_{W,z}} \cdot \frac{\sigma_W}{\sigma_z} \cdot z + \xi \quad \xi \sim \mathcal{N}(0, \sigma_\xi^2) \quad \xi \perp z$$

$$\underbrace{\mathbb{E} \{ W \cdot z \}}_{\text{Cov}(W,z)} = \underbrace{\rho_{W,z} \cdot \frac{\sigma_W}{\sigma_z} \cdot \sigma_z^2}_{\text{Cov}(W,z)} + \mathbb{E} \{ \xi \cdot z \}$$

$$\Rightarrow \mathbb{E} \{ \xi \cdot z \} = 0 \quad \Rightarrow \quad \begin{aligned} &\xi \perp z \\ &\xi, z \text{ joint Gaussian} \\ &\text{both centered} \\ &\text{Cov} = 0 \text{ if joint Gaussian} \\ &\Rightarrow \xi \perp z \end{aligned}$$

$$\Rightarrow \mathbb{E} \{ W f(z) \} = \mathbb{E} \left\{ \left(\rho \frac{\sigma_W}{\sigma_z} z + \xi \right) f(z) \right\}$$

$$= \underbrace{\rho \frac{\sigma_W}{\sigma_z}}_{\text{Cov}(W,z)} \underbrace{\mathbb{E} \{ z \cdot f(z) \}}_{\sigma_z^2 \mathbb{E} \{ f'(z) \}} + \underbrace{\mathbb{E} \{ \xi f(z) \}}_{\xi \perp z}$$

$$= \underbrace{\rho \sigma_W \sigma_z}_{\text{Cov}(W,z)} \mathbb{E} \{ f'(z) \}.$$

Extension to D-dim.

(*) Goal: $A = \mathbb{E} \{ Z_i f(\underline{z}) \}$
 $= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} z_i f(\underline{z}) \Phi(\underline{z}) d\underline{z}$

$$\Phi(\underline{z}) = \underbrace{\frac{1}{(\sqrt{2\pi})^d \sqrt{\det(\Sigma)}}}_C \exp \left\{ -\frac{1}{2} \underline{z}^T \Sigma^{-1} \underline{z} \right\}$$

$$\nabla_{\underline{z}} \Phi = C \cdot \exp \left\{ -\frac{1}{2} \underline{z}^T \Sigma^{-1} \underline{z} \right\} \cdot (-\Sigma^{-1} \underline{z})$$

$$\Sigma \nabla_{\underline{z}} \Phi = -\Phi(\underline{z}) \cdot \underline{z}$$

$$z_i \Phi(\underline{z}) = -\Sigma_{i,j} \nabla_j \Phi$$

$$= -\sum_{j=1}^d \Sigma_{i,j} \underbrace{\partial_j \Phi}_{\frac{\partial \Phi}{\partial z_j}}$$

$$A = \int -\sum_{j=1}^d \Sigma_{i,j} \partial_j \Phi(\underline{z}) \cdot f(\underline{z}) d\underline{z}$$

$$= -\sum_{j=1}^d \int \Sigma_{i,j} \partial_j \Phi(\underline{z}) \cdot f(\underline{z}) d\underline{z}$$

$$= \sum_{j=1}^d \Sigma_{i,j} \int \partial_j f(\underline{z}) \Phi(\underline{z}) d\underline{z}$$

$$= \sum_{j=1}^d \Sigma_{i,j} \mathbb{E} \{ \partial_j f(\underline{z}) \}$$

Recall:

$$\int \partial_j \Phi(\underline{z}) \cdot f(\underline{z}) d\underline{z}$$

$$= - \int \Phi(\underline{z}) \cdot \partial_j f(\underline{z}) d\underline{z}$$

In the special case $f(\underline{z}) = \partial_k F(\underline{z})$

$$\Rightarrow A = \sum_{j=1}^d \Sigma_{i,j} \mathbb{E} \{ \partial_{j,k}^2 F(\underline{z}) \}$$

$$F_{\beta}(X_1, \dots, X_n) := \frac{1}{\beta} \log \left(\sum_{i=1}^N \exp(\beta X_i) \right)$$

$$\lim_{\beta \rightarrow \infty} F_{\beta}(X_1, \dots, X_n) = \max_{1 \leq i \leq N} X_i$$

It suffices to show that

$$\forall \beta \quad \mathbb{E} \{ F_\beta(x_1, \dots, x_n) \} \leq \mathbb{E} \{ F_\beta(Y_1, \dots, Y_n) \}$$

to show that $\mathbb{E} \left\{ \max_{1 \leq i \leq N} x_i \right\} \leq \mathbb{E} \left\{ \max_{1 \leq i \leq N} Y_i \right\}.$

Define $\underline{Z}(t) = \sqrt{t} \underline{Y} + \sqrt{1-t} \underline{X}$

$$\Psi(t) = \mathbb{E} \{ F_\beta(\underline{Z}_1(t), \dots, \underline{Z}_N(t)) \} \quad \Psi(0) \stackrel{\text{IE}}{=} \mathbb{E} \{ F_\beta(x_1, \dots, x_N) \}$$

$$\Psi(1) = \mathbb{E} \{ F_\beta(Y_1, \dots, Y_N) \}$$

$$\forall \beta > 0, \quad \Psi'(t) \geq 0 \quad 0 \leq t \leq 1 \quad \text{it's all done!}$$

So let's compute $\Psi'(t)$ then!

$$\Psi'(t) = \mathbb{E} \left\{ \sum_{i=1}^N \partial_i F_\beta \underbrace{\frac{dZ_i(t)}{dt}}_{\frac{1}{2\sqrt{t}} Y_i - \frac{1}{2\sqrt{1-t}} X_i} \right\}$$

$$= \sum_{i=1}^N \mathbb{E} \left\{ \frac{1}{2\sqrt{t}} Y_i \partial_i F_\beta(\underline{Z}(t)) - \frac{1}{2\sqrt{1-t}} X_i \partial_i F_\beta(\underline{Z}(t)) \right\}$$

$$= \sum_{i=1}^N \sum_{j=1}^N \frac{1}{2\sqrt{t}} \text{Cov}(Y_i, Z_j) \partial_{i,j}^2 F_\beta - \frac{1}{2\sqrt{1-t}} \text{Cov}(X_i, Z_j) \partial_{i,j}^2 F_\beta$$

$$\text{Cov}(Y_i, \sqrt{t} Y_j + \sqrt{1-t} X_j) = \sqrt{t} \mathbb{E}(Y_i Y_j)$$

$$\text{Cov}(X_i, \sqrt{t} Y_j + \sqrt{1-t} X_j) = \sqrt{1-t} \mathbb{E}(X_i X_j)$$

$$\partial_i F_\beta = \partial_i \left(\frac{1}{\beta} \log \sum \exp(\beta z_i) \right)$$

$$= \frac{\frac{1}{\beta} \beta \exp(\beta z_i)}{\sum_{j=1}^N \exp(\beta z_j)} = \frac{\exp(\beta z_i)}{\sum_{j=1}^N \exp(\beta z_j)} =: p_i(\underline{z})$$

$$\partial_{ij}^2 F_\beta = \delta_{ij} \left[\frac{\beta \exp(\beta z_i) \sum_{j=1}^N \exp(\beta z_j) - \beta \exp(2\beta z_i)}{\left(\sum_{j=1}^N \exp(\beta z_j) \right)^2} \right]$$

$$+ (1 - \delta_{ij}) \left[\frac{-\beta \exp(\beta z_j) \exp(\beta z_i)}{\left(\sum_{j=1}^N \exp(\beta z_j) \right)^2} \right]$$

$$= \delta_{ij} \beta p_i - \beta p_i p_j = \beta (\delta_{ij} p_i - p_i p_j)$$

$$\psi'(t) = \frac{1}{2} \sum_{i,j} \underbrace{(\mathbb{E}\{Y_i Y_j\} - \mathbb{E}\{X_i X_j\})}_{:= a_{ij}} (\beta [\delta_{ij} p_i - p_i p_j])$$

$$= \frac{1}{2} \sum_{i,j} a_{ij} \beta [p_i \delta_{ij} - p_i p_j]$$

$$\text{Claim} = \frac{\beta}{4} \sum_{i,j} \underbrace{[a_{ii} + a_{jj} - 2a_{ij}]}_{\mathbb{E}((Y_i - Y_j)^2) - \mathbb{E}((X_i - X_j)^2) \geq 0} p_i p_j$$

$$\mathbb{E}((Y_i - Y_j)^2) - \mathbb{E}((X_i - X_j)^2) \geq 0$$